

A disproof of conjecture 00000002601:
the maximum number of fixed points of a monotone self-map
of a finite lattice is the lattice size, not the chain length

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Abstract

Conjecture 00000002601 asserts that the maximum number of fixed points of a monotone self-map of an n -element finite lattice equals the length of a maximal chain, that this maximum is attained on distributive lattices, and that on non-distributive lattices the maximum drops strictly, the drop being the minimal number of embedded N_5 sublattices. We show the conjecture is false. The identity map is monotone and fixes all n elements, so the maximum is exactly n — the trivial upper bound — on every finite lattice. On the four-element distributive lattice B_2 the maximum is 4, while a maximal chain $\perp < a < \top$ has 3 elements (equivalently 2 edges). The claimed equality therefore fails already on a distributive lattice, and it fails under both readings of “chain length”. The strict drop on non-distributive lattices is likewise false: on N_5 the identity still fixes all 5 elements, so the drop is 0. All claims are verified by exhaustive enumeration and, independently, in core Lean 4.

1 The conjecture and its refutation

Definition 1 (monotone self-map, fixed-point count). Let L be a finite lattice with order $\leq$. A self-map $f: L \rightarrow L$ is *monotone* (order-preserving) if $x \leq y$ implies $f(x) \leq f(y)$ for all $x, y \in L$. Its fixed-point set is $\text{Fix}(f) = \{x \in L : f(x) = x\}$.

The conjecture under consideration is the following.

Conjecture 00000002601. *The maximum number of fixed points of a monotone map on an n -element finite lattice equals the length of a maximal chain, attained on distributive lattices; the maximum drops strictly on non-distributive lattices, the drop being the minimal number of embedded N_5 sublattices.*

The length of a chain admits two conventions: the number of its elements, or the number of its edges (covering pairs). The conjecture is ambiguous between them; we refute it under both. The main observation is utterly elementary.

Theorem 2 (identity bound). *For every finite lattice L with $|L| = n$, the identity map id_L is monotone and satisfies $\text{Fix}(\text{id}_L) = L$, so $|\text{Fix}(\text{id}_L)| = n$. Since no self-map of L can fix more than n elements, the maximum number of fixed points over all monotone self-maps of L is exactly n , the trivial upper bound.*

Proof. For $x \leq y$ we have $\text{id}_L(x) = x \leq y = \text{id}_L(y)$, so id_L is monotone. Every element is fixed, hence $|\text{Fix}(\text{id}_L)| = n$. The number of fixed points of any self-map is at most $|L| = n$. Therefore the maximum equals n . $\square$

Consequently the quantity the conjecture tries to compute is trivially n on *every* finite lattice, distributive or not. The conjectured equality with the length of a maximal chain can only hold when a maximal chain has n elements (equivalently $n - 1$ edges), i.e. when the lattice is itself a chain. As soon as L is not a chain, some element lies off every maximal chain and the equality fails. This already happens for the simplest non-chain distributive lattice.

Corollary 3. *Let L be a finite lattice of size n that is not a chain. Then the maximum number of fixed points of a monotone self-map of L is n , which is strictly larger than the number of elements of a maximal chain and strictly larger than its number of edges. In particular Conjecture 00000002601 is false for every finite lattice that is not a chain, and its “attained on distributive lattices” rider does not rescue it.*

Proof. By Theorem 2 the maximum is n . A maximal chain of L has at most $n - 1$ elements, since L is not a chain, and therefore at most $n - 2$ edges; both are strictly less than n . $\square$

2 The explicit counterexample B_2

Let $B_2 = \{0, a, b, 1\}$ be the Boolean lattice on two atoms, i.e. the four-element distributive lattice with 0 bottom, 1 top, and a, b incomparable. Its Hasse diagram is the square $0 < a < 1, 0 < b < 1$.

Example 4. On B_2 :

- the identity map is monotone and has 4 fixed points;
- a maximal chain is $0 < a < 1$, which has 3 elements (2 edges);
- hence the maximum number of fixed points (4) differs from the maximal chain length under either convention: $4 \neq 3$ and $4 \neq 2$.

The seven non-identity elements of the lattice of all 36 monotone self-maps of B_2 fix at most 3 elements, so the maximum 4 is genuinely attained by the identity and not by an accident of a degenerate map. In particular the failure of the conjectured equality is not repaired by excluding the identity: on B_3 (below) a non-identity monotone map still fixes 7 elements, which exceeds the maximal chain length 4.

3 Exhaustive computations

We verified the relevant maxima by brute force. For completeness we record the exact counts; all of them are reproduced by `reproduce.py` and, for B_2 , by the Lean development in `lean4/`.

3.1 B_2 : all $4^4 = 256$ self-maps

Enumerating all 256 self-maps of B_2 and retaining the monotone ones gives exactly 36 monotone maps. The maximum of $|\text{Fix}(f)|$ over them is 4, attained by the identity; the maximum over non-identity monotone maps is 3. The maximal chain has 3 elements and 2 edges. Hence

$$4 = \max_f |\text{Fix}(f)| \neq 3 = \#\{\text{elements of a maximal chain}\},$$

and likewise $4 \neq 2$ for the edge count.

3.2 B_3 : all $8^8 = 16\,777\,216$ self-maps, enumerated by pruning

Let B_3 be the Boolean lattice on three atoms, of size 8. A naive scan of all $8^8 = 16\,777\,216$ self-maps is possible in principle but wasteful; the script instead enumerates the monotone maps directly by depth-first search, assigning the image of each element in a topological order and pruning any partial assignment that violates monotonicity. This enumerates exactly the monotone self-maps and finds

$$\#\{\text{monotone self-maps of } B_3\} = 8000,$$

with maximum fixed-point count 8 (the identity) and maximum over non-identity monotone maps 7. A maximal chain of B_3 has 4 elements (3 edges). Thus $8 \neq 4$ and $8 \neq 3$.

Remark 5 (non-identity maps). On B_3 , the monotone map that sends one coatom to the top element and fixes everything else is not the identity and fixes 7 elements. Since $7 > 4$ and $7 > 3$, excluding the identity does not restore the conjecture.

4 The non-distributive lattice N_5

Let N_5 be the five-element non-distributive lattice with bottom 0, top 1, and middle elements a, b, c satisfying

$$0 < a < b < 1, \quad 0 < a < c < 1,$$

with b and c incomparable. The identity is monotone and fixes all 5 elements, so the maximum number of fixed points is 5, while a maximal chain $0 < a < b < 1$ has 4 elements (3 edges). Consequently

$$\max_f |\text{Fix}(f)| = 5 \neq 4 \quad \text{and} \quad 5 \neq 3.$$

More importantly for the second half of the conjecture: the maximum on N_5 is $5 = |N_5|$, exactly the same trivial value as on *any* five-element lattice, distributive or not. So the maximum does *not* drop strictly on the non-distributive lattice N_5 ; the drop is 0. Any clause tying a strict drop to non-distributivity, or to the number of embedded N_5 sublattices, is therefore false as stated: the conjectured quantity is blind to distributivity. (For reference, exhaustive enumeration shows N_5 has 133 monotone self-maps, the maximum among non-identity ones being 4; but the identity alone already realizes the true maximum 5.)

5 The elements-versus-edges ambiguity

The conjecture says “the length of a maximal chain” without specifying whether a chain of k elements has length k or $k - 1$. We stress that the refutation is independent of this choice:

Lattice	$\max_f \text{Fix}(f) $	chain elements	chain edges	refuted?
B_2 (distr., $n = 4$)	4	3	2	yes (both)
B_3 (distr., $n = 8$)	8	4	3	yes (both)
N_5 (non-distr., $n = 5$)	5	4	3	yes (both)

In every case the maximum number of fixed points equals the lattice size n (Theorem 2) and is strictly larger than both candidate chain lengths. The conjectured equality fails on the distributive lattices B_2 and B_3 as well, so the “attained on distributive lattices” part is false too.

6 Formal verification

The file `lean4/Main.lean` formalizes the B_2 refutation in core Lean 4 (`import Std`, no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`). It encodes B_2 as `Fin 4`, defines monotonicity, proves that the identity is monotone and fixes all four elements, and proves by exhaustive computation over the 256 functions that

$$\text{maxFixedPoints} = 4, \quad \text{maxChainLen} = 3, \quad 4 > 3,$$

culminating in `conjecture_00000002601_false`. A companion `Check.lean` prints the axioms of every theorem; none depends on `sorryAx` or `ofReduceBool`. The script `reproduce.py` independently reproduces all numerical claims above (including B_3 and N_5) using only the Python standard library.

References

- [1] A. Tarski, *A lattice-theoretical fixpoint theorem and its applications*, Pacific J. Math. **5** (1955), 285–309.
- [2] G. Grätzer, *Lattice Theory: Foundation*, Birkhäuser, 2011.